Encapsulation of Fungi To Enhance Bioaugmentation Systems for PAH Remediation

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Background

Polycyclic aromatic hydrocarbon (PAHs) are a class of carcinogenic compounds that are very persistent in the environment.[1-3] Traditional strategies to remediate them are costly and ecologically disruptive.[4]

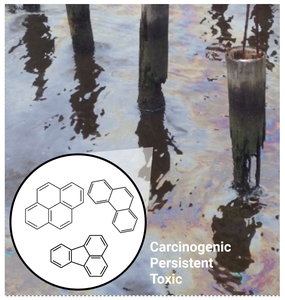
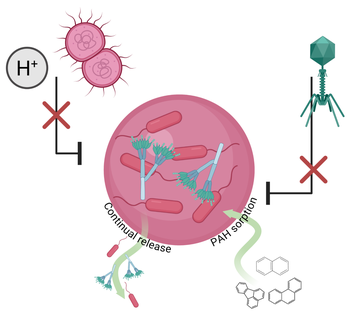


Fig 1. Elizabeth River with visible PAH contamination with example model PAHs

Bioaugmentation is a promising strategy to that **adds** microbes to the contaminated environment to metabolize the PAHs. **However, studies show minimal success under environmentally relevant conditions.** [5]



We are encapsulating the microbes in **sodium alginate** to protect them from environmental stressors, promote establishment through continual release, and act as a sink for PAHs. [6]

Fig 2. Schematic of the encapsulated microbes showing some of the expected effects

Project Goals

1. **Assess growth and viability of encapsulated fungal isolates**
2. **Quantify apparent PAH removal (liquid media and the dissolved capsule)**
3. **Measure apparent PAH removal in encapsulated consortia**

Methods

Sodium alginate microbial mixtures are extruded into 0.1 M calcium chloride to produced stable porous capsules. [7]

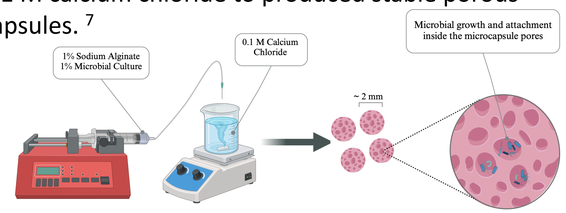


Fig 3. Schematic of the encapsulated microbes showing some of the expected effects

Results

Long Term Viability

Encapsulation **promoted viability of Trichoderma** cultures after 8 weeks but had **no effect on Aureobasidium**

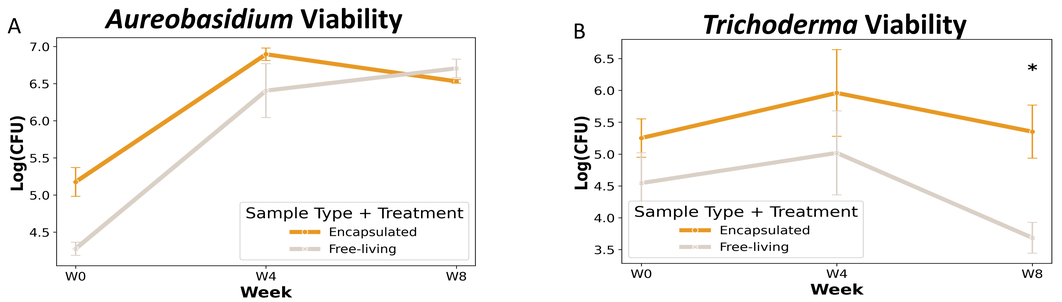
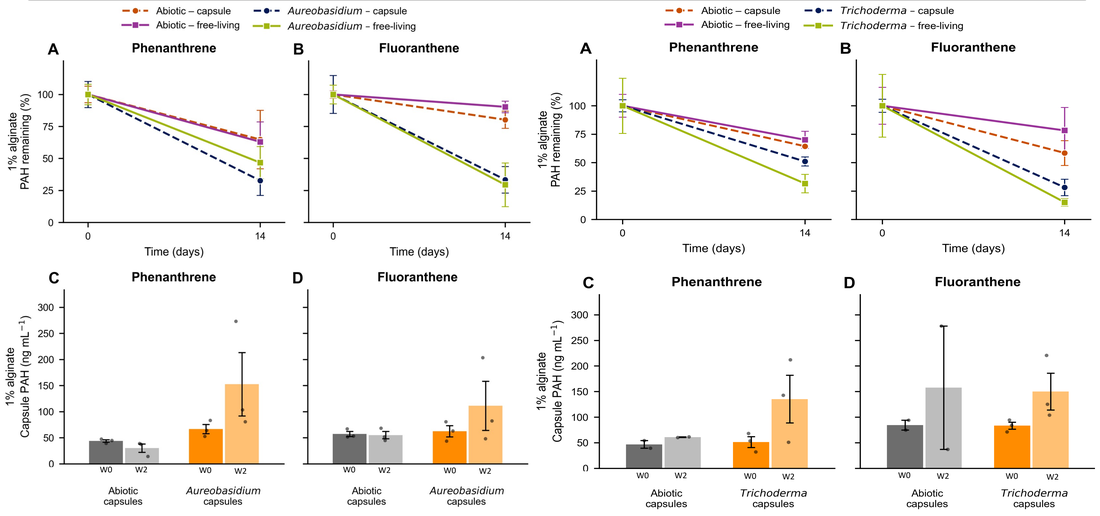


Fig 3 (A-B). Long-term viability of encapsulated and free-living fungal cultures over 8 weeks. Colony-forming units were measured at week 0,4, and 8 for *Aureobasidium* (A) and *Trichoderma* (B). Capsules consisted of 1% alginate extruded at 18 mL h^{-1}. Error bars represent standard error from the mean (n=3).

Apparent removal

PAH **accumulated on capsules** over 2 weeks but showed similar PAH removal trends to free-living cultures.



PAH recovery and capsule-associated retention in **Fig 5 (A-D).** *Aureobasidium* (left) and **Fig 6 (A-D).** *Trichoderma* (right) treatments over 14 days. Columns show phenanthrene and fluoranthene where (A–B) show the percentage of PAH mass remaining relative to day 0 under different conditions and (C-D) (D–F) measure the capsule-associated PAH concentrations recovered from 1% alginate capsules at days 0 and 14.

Consortia effects on capsule sorption

Bacterial capsules **promoted removal of capsule-bound PAHs** whereas fungal capsules **showed an accumulation** over 8 weeks

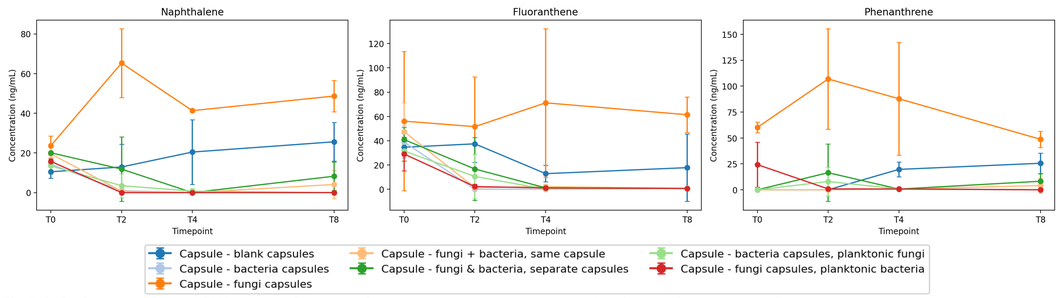


Fig 7 (A-C). Concentrations of PAHs over an 8 week period on the capusles only between fungi, bacteria, or a consortia.

Conclusions

In lab-controlled conditions

- Encapsulated Trichoderma showed increased viability
- Capsules accumulated PAHs
- Bacteria drove PAH removal in encapsulated consortia

Future Directions

- Use contaminated sediments to assess the capsules' effect under environmentally relevant conditions
- Create new capsule formulations to increase the bioavailability of PAHs in the environment

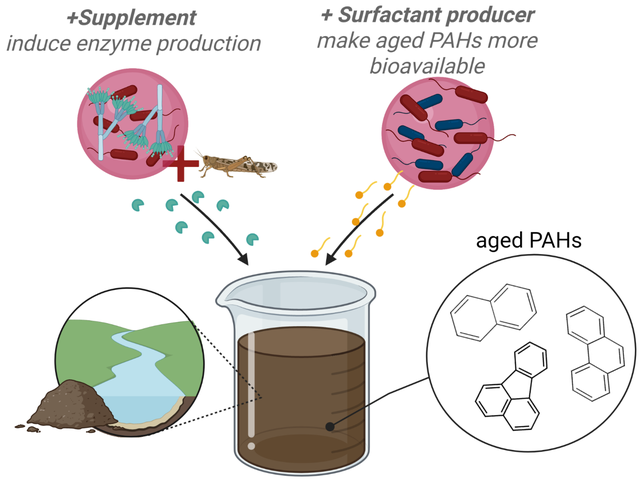


Fig 8 (A-B). Schematic of future experiments incorporating PAH contaminated sediments and capsule formulations hypothesized to help improve removal of aged PAHs

References and Acknowledgments

[1] US EPA (2023), Superfund [2] Di Giulio, R. T., & Clark, B. W. (2015). *J Tox and Env Health* [3] Baek, et al. (1991) *Water Air Soil Pollut*. [4] Seo, J.-S., Keum, Y.-S., & Li, Q. X. (2009). Bacterial Degradation of Aromatic Compounds. [5] Omokhagbor Adams et al., (2020) *International Journal of Environmental Bioremediation & Biodegradation* [6] Babusci & Gunsch. (Under Review). *Journal of Hazardous Materials Advances* [8] Yueng, et al. (2016) *Front. Microbial.*

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